

Food Inflation Forecasting - Feature Engineering Report

1. Dataset Integration

Five datasets were integrated into a unified monthly food inflation forecasting dataset. Datasets included: - Baltic Dry Index shipping indicators - Brent crude oil prices - Botswana policy rate - FAO food price indicators - Human Capital Project indicators All datasets were aligned using monthly timestamps.

2. Data Transformation

Raw datasets were cleaned and converted into machine learning ready format. Long format datasets were transformed into wide format using country-indicator feature creation.

3. Feature Engineering

The following features were created: - Monthly and quarterly time features - Lag features (1, 3, 6 and 12 months) - Rolling averages - Economic change indicators - Baltic Dry Index volatility features These features allow the model to capture trends, seasonality and delayed economic effects.

4. Final Dataset

The final machine learning dataset contains 288 monthly observations from January 2000 to December 2023. The target variable is FAO_CP_23012 food price index.